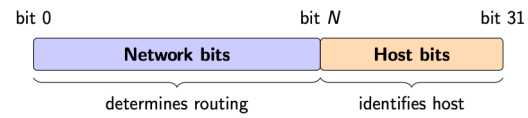
Lecture 14 - IP Addressing, CIDR, Subnetting

How addresses are structured, allocated, and divided

Acronym	Meaning
APIPA	Automatic Private IP Addressing
ARIN	Am. Registry for Internet Numbers
CGN	Carrier-Grade NAT
CIDR	Classless Inter-Domain Routing
DHCP	Dynamic Host Configuration Protocol
IANA	Internet Assigned Numbers Authority

Acronym	Meaning
NAT	Network Address Translation
RFC	Request for Comments
RIR	Regional Internet Registry
RIPE	Réseaux IP Européens NCC
VLSM	Variable-Length Subnet Masking
VPC	Virtual Private Cloud

Core mental model: one address, two jobs



IPv4 → 32 bit that encode 2 peices of information:

Internet core: `128.197.0.0/16` → "toward BU"

BU edge router: `128.197.10.0/24` → "toward Photonics"

Building switch: `128.197.10.42` → port 7

Hierarchical routing:

network portion (identifies the network) and a host portion (identifies a specific device)

IPv4: 32 bits, dotted decimal notation

32 bit address space → $2^{32} \approx 4.3$ billion addresses total

Dotted decimal → 32 bits split into 4, 8 bit octects → each between 0 -255

```
Binary: 10000000.11000101.00001010.00101010
Dotted-decimal: 128 .197 .10 .42
```

Key conversions

Decimal	Binary	Decimal	Binary
0	0000 0000	192	1100 0000
128	1000 0000	224	1110 0000
255	1111 1111	240	1111 0000

These come up constantly in masks.

Classful addressing

Class	Bits	Prefix	Hosts/net	Range
A	0	/8	16.7M	0.0.0.0 – 127.x.x.x
B	10	/16	65,534	128.x.x.x – 191.x.x.x
C	110	/24	254	192.x.x.x – 223.x.x.x
D	1110	—	multicast	224.x.x.x – 239.x.x.x
E	1111	—	reserved	240.x.x.x – 255.x.x.x

--> reading an address:

Class A lottery

Before, organization could only apply for class A → 16.7 million addresses
Now → 650 million dollars

Organization	Block	Status
US Dept. of Defense	6, 7, 11, 21, 22, 26, . . .	Retained (~13 /8s)
IBM	9.0.0.0/8	Retained
AT&T	12.0.0.0/8	Retained
Apple	17.0.0.0/8	Retained
MIT	18.0.0.0/8	Partial — sold half to Amazon (2017, ∼$4M)
Ford Motor Company	19.0.0.0/8	Retained
General Electric	3.0.0.0/8	Transferred → Amazon AWS

Why classful addressing failed → 1.address waste

The binary choice → Class B (65K hosts) or Class C (254 hosts) → matched almost no real organization's needs → roughly half of all assigned IPv4 address space sat unused inside organizations that had been given overly large blocks

Organization	Hosts needed	Class assigned	Addresses wasted
Small company (50 hosts)	50	C (/24, 254)	~200 manageable
Medium company (2K hosts)	2000	B (/16, 65K)	63,000
BU (5K students)	5000	B (/16, 65K)	60,000
Large ISP (1M hosts)	1M+	A (/8, 16.7M)	15,700,000

Address Class	1st Octet range in decimal	1st Octet bits (Blue Dots do not change)	Network (N) and Host (H) Portion	Default mask (Decimal)	Number of possible networks and hosts per network
A	0–127	00000000 - 01111111	N.H.H.H	255.0.0.0	128 Nets (2^7) 16,777,214 hosts (2^{24}–2)
B	128–191	10000000 - 10111111	N.N.H.H	255.255.0.0	16,384 Nets (2^{14}) 65,534 hosts (2^{16}–2)
C	192–223	11000000 - 11011111	N.N.N.H	255.255.255.0	2,09,150 Nets (2^{21}) 254 hosts (2^8–2)
D	224–239	11100000 - 11101111	NA (Multicast)	-	-
E	240–255	11110000 - 11111111	NA (Experimental)	-	-

MORE VIDEOS cademy.org

Why classful addressing failed → 2.routing explosion

A medium company needing 2000 hosts but denied a Class B received 8 separate Class C blocks. Each required its own entry in the global routing table — no aggregation was possible under classful rules.

The core mismatch

Class B = 65K hosts. Class C = 254 hosts. Medium organizations fell into neither category — they needed multiple Class C blocks that could not be aggregated.

CIDR: the fix → classless inter domain routing made 3 changes:

1. Any prefix length is valid
2. Prefix length is written explicitly
3. Aggregation is first class

Before CIDR — 8 separate routing entries:

192.168.H.0/24
192.168.G.0/24
192.168.F.0/24
192.168.E.0/24
192.168.D.0/24
192.168.C.0/24
192.168.B.0/24
192.168.A.0/24

After CIDR — 1 aggregated entry:

192.168.0.0/21

covers all 8 /24s (2048 addresses)

Subnet masks and prefix lengths

A CIDR prefix is equivalent to a subnet mask: N ones followed by (32 − N) zeros.

```
/24 -> 11111111.11111111.11111111.00000000 -> 255.255.255.0
/20 -> 11111111.11111111.11110000.00000000 -> 255.255.240.0
/26 -> 11111111.11111111.11111111.11000000 -> 255.255.255.192
/30 -> 11111111.11111111.11111111.11111100 -> 255.255.255.252
```

Prefix	Total addresses	Usable hosts	Typical use
/30	4	2	Point-to-point links
/26	64	62	Small office
/24	256	254	Standard LAN segment
/22	1,024	1,022	Large campus segment
/16	65,536	65,534	Large organization

→ useable hosts → $2^{32-N}-2$

Network address, broadcast, host range

Given address and prefix → compute:

1. Network address → address and mask (all host bits zeroed)

```
  192.168.1.75 = 11000000.10101000.00000001.01001011
& 255.255.255.0 = 11111111.11111111.11111111.00000000
                  ---------------------------------------
                  11000000.10101000.00000001.00000000 = 192.168.1.0
```

2. broadcast address → all host bits set to 1

`192.168.1.0/24` → broadcast = `192.168.1.255`

3. Usable host rane → network + 1 through broadcast address - 1

`192.168.1.0/24` → hosts `192.168.1.1` – `192.168.1.254` (254 hosts)

The alignment rule

A valid subnet's network address must be a **multiple of the block size**. For a /26 (block size 64): valid network addresses are .0, .64, .128, .192. `192.168.1.10/26` is **invalid** — .10 is not a multiple of 64.

subnetting : dividing an address block

Subnet → extend the prefix to create smaller blocks

Rule: borrow k bits ⇒ 2^k subnets, each a /(N+k)

Subnetting: the bit level view

Alignment rule

Each subnet's network address must be a **multiple of the block size**. For /26 (block size 64): valid starts are .0, .64, .128, .192 only. `192.168.10.10/26` is invalid — .10 is not a multiple of 64.

Subnet membership check

Are 2 hosts in the same subnet ? → AND each address with the mask and compare

192.168.10.75 and **192.168.10.100** — same /26?

```
.75 AND 255.255.255.192 -> 192.168.10.64
.100 AND 255.255.255.192 -> 192.168.10.64 <- same network -> same subnet
```

192.168.10.75 and **192.168.10.130** — same /26?

```
.75 AND 255.255.255.192 -> 192.168.10.64
.130 AND 255.255.255.192 -> 192.168.10.128 <- different -> must route
```

Key implication

Hosts in different subnets cannot communicate directly — traffic between them must pass through a **router**, even if both hosts are on the same floor of the same building. Subnet boundaries are logical, not physical.

Variable length subnet masking (VLSM)

Subnets within an orgainziation done need to be equal sized. VLSM allows different prefix lengths in different parts of the address space

Avoid overlaps

VLSM is efficient but requires careful bookkeeping. Always verify that no two assigned blocks overlap. The alignment rule keeps things tidy: each block must start at a multiple of its block size.

RFC 1918: private address space → designated three blocks as private address space — never assigned to hosts on the public internet

Block	Range	Size	Typical use
`10.0.0.0/8`	10.0.0.0 – 10.255.255.255	16M	Enterprise, cloud VPCs
`172.16.0.0/12`	172.16.0.0 – 172.31.255.255	1M	Medium enterprise
`192.168.0.0/16`	192.168.0.0 – 192.168.255.255	65K	Home networks, small offices

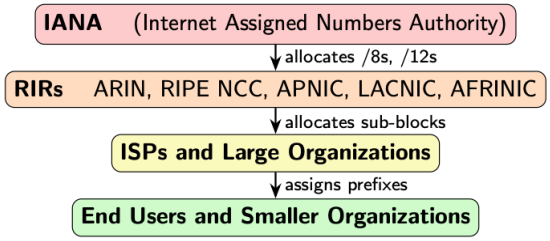
Private addresses are:

- Not routable on the public internet
- Reusable
- Used behind NAT

Other special purpose ranges

Range	Purpose	Notes
`127.0.0.0/8`	Loopback	`127.0.0.1` = localhost; packets never leave host
`169.254.0.0/16`	Link-local (APIPA)	Auto-assigned when DHCP fails; not routable beyond link
`224.0.0.0/4`	Multicast	OSPF: 224.0.0.5/6; RIP: 224.0.0.9
`255.255.255.255`	Limited broadcast	All hosts on local subnet; not forwarded by routers
`0.0.0.0`	"This host"	Used in DHCP Discover before address is assigned
`100.64.0.0/10`	Carrier-Grade NAT	RFC 6598; ISP-internal; not on public internet

The allocation hierarchy → IPv4 addresses are allocated through a 4 level hierarchy



ARIN → noth america
RIPE NNC → europe, middle east, central asia
APNIC → asia pacific
LACNIC → latin america and caribbean
AFRINIC → africa

IPv4 Exaustion

- $2^{32} \approx 4.3$ billion addresses → the internet grew faster
- New IPv4 addresses are no longer available from RIRs. They are bought on the transfer market at $30–50 per address ($8K–12K for a /24)
- One public IP representing thousands of private hosts extended IPv4 life by ~20 years beyond original projections.

Wireshark: spot private vs. public addresses in browser traffic